

Complete Superconvergence Phases and Optimal Phase Filters for Rvachev Quadrature

A quantitative completion of a ProveIt research frontier

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Abstract

The ProveIt Fabius-function project establishes polynomial quadrature exactness at parity-selected lattice phases, while explicitly leaving open whether other first-superconvergent phases exist. We prove that they do not. For every positive integer mesh parameter M , write $d = v_2(M)$. Exactly two phases modulo one integrate all polynomials through degree $d + 1$: $0, \frac{1}{2}$ when d is even, and $\frac{1}{4}, \frac{3}{4}$ when d is odd. The proof is quantitative. An odd-multiplier inequality for the Fourier product makes the entire first defect a relative perturbation of one harmonic by at most $\pi^2/8 - 1 < 0.234$, uniformly in the odd part of M . We also classify every finite complex phase measure giving polynomial exactness uniformly under translation. At a reduced rational mesh a/b , exactness through degree r is equivalent to invariance under translation by $1/(b2^{\max(0, r - v_2(a))})$. At an irrational mesh, even uniform constant exactness forces Haar measure. The classification gives sharp phase-count bounds, unique minimum-size filters up to translation, a finite Fourier-moment recognition theorem, and corresponding results for moment-deconvolved polynomial synthesis. Complete analytic proofs, exact sample checks, and a further-research program are included.

Status. The results below are proved in ordinary mathematics; no new Lean or Rocq verification is claimed. Their novelty is assessed relative to the specifically inspected repository statements and primary references, not certified against the entire literature. This is an unrefereed research manuscript, not a claim to have solved a major named conjecture.

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1 The question and the precise contribution

A lattice quadrature can have greater polynomial exactness at a special translation than at a generic translation. The central question here is whether the familiar symmetry-selected translations of Rvachev quadrature are the *only* ones. A second question asks how efficiently several translated copies can be combined to obtain exactness at *every* translation. These are different problems, and keeping their quantifiers separate is essential.

Let u be the normalized Rvachev up-function, supported on $[-1, 1]$ and satisfying $\int_{\mathbb{R}} u = 1$. For a mesh parameter $M > 0$ and a phase $\theta \in \mathbb{T} = \mathbb{R}/\mathbb{Z}$, put

$$Q_{M,\theta}(P) = \frac{1}{M} \sum_{k \in \mathbb{Z}} P\left(\frac{k+\theta}{M}\right) u\left(\frac{k+\theta}{M}\right), \quad I(P) = \int_{\mathbb{R}} P(x)u(x) dx. \quad (1.1)$$

The actual lattice spacing is $1/M$. The sum is finite because u is compactly supported. We write $\mathcal{P}_r$ for the complex polynomials of degree at most r , including the zero polynomial.

The source module `RvachevSuperconvergentSynthesis.lean` proves that, for $M \in \mathbb{N}_{>0}$, every phase is exact on $\mathcal{P}_{v_2(M)}$, and the indicated parity-selected phases are exact on $\mathcal{P}_{v_2(M)+1}$. Its introductory documentation explicitly disclaims completeness of that phase list. The module `CombDefectSeries.lean` supplies the absolutely convergent Fourier series for the defect and its odd-frequency support at the first failing degree [4, 5].

The repository reference used here is the commit

de4ac5ab1438bcbeed58481c4449b593074a1a1f.

We inspected the relevant module statements, proof portions, and frontier overview, not every research artifact in the repository. The classical function, its Fourier-product description, and its dyadic arithmetic are not new; primary references include Arias de Reyna [1, 2]. The use of Fourier zeros to control polynomial reproduction belongs to the Strang–Fix tradition; see, for example, Zakharov [3]. Our contribution is the specific quantitative phase-completeness argument and the full phase-measure classification worked out for this kernel.

1.1 Results at a glance

Theorem 4.1 proves the missing necessity statement, for *all* positive integer M , not just powers of two. It also gives a uniform relative error bound, the sign of the first defect, and simplicity of its two zeros. The key ingredient is the estimate

$$\left| H\left(\frac{ak}{2}\right) \right| \leq \frac{1}{k^2} \left| H\left(\frac{a}{2}\right) \right| \quad (a, k \text{ positive odd}),$$

where $H = \widehat{u}$. The estimate retains the odd part a of the mesh; it does not replace it by a mesh-dependent unspecified constant.

Theorem 5.4 identifies every mass-one finite complex measure on the phase circle that is uniformly exact through a prescribed degree. The answer is an exact cyclic-invariance condition at rational meshes and a Haar-measure rigidity statement at irrational meshes. Theorems 6.3 and 6.4 then solve the finite phase-budget problem, including its equality cases. Negative or complex weights do not improve the bound. Theorem 7.1 turns cyclic invariance of a real signed atomic filter into a finite set of Fourier-moment tests.

Finally, Theorem 8.1 transfers the results to polynomial synthesis after moment deconvolution. Exact rational computations and numerical product checks provide independent diagnostics, not substitutes for the proofs.

1.2 What the statements do not assert

The phase theorem classifies exactness through the *first additional* degree $\nu_2(M) + 1$. It does not prove that every selected phase necessarily fails at the next degree $\nu_2(M) + 2$. That separate fixed-phase maximality problem remains in the research program. Conversely, our finite-filter maximality theorem is complete, but concerns exactness *uniformly in the translating phase*. It is not a lower bound for unrestricted Gaussian quadrature or for arbitrary node placement at a single fixed phase.

2 Normalization, Fourier zeros, and the defect series

2.1 A construction fixing all factors of two

Take independent random variables V_j , each uniform on $[-1, 1]$, and set

$$Y = \sum_{j=1}^{\infty} 2^{-j} V_j. \quad (2.1)$$

The sum converges absolutely and lies in $[-1, 1]$. With the convention

$$\widehat{f}(t) = \int_{\mathbb{R}} f(x) e^{-2\pi i t x} dx, \quad \text{sinc } z = \begin{cases} \sin z/z, & z \neq 0, \\ 1, & z = 0, \end{cases}$$

its characteristic function is

$$H(t) = \prod_{j=0}^{\infty} \text{sinc}\left(\frac{\pi t}{2^j}\right). \quad (2.2)$$

This choice of Y is important: starting the product at $j = 1$ instead would change the support and every mesh convention.

For every compact subset of $\mathbb{C}$, the tail factors differ from 1 by $O(4^{-j})$, uniformly on that subset. Hence the product defines an entire function. On the real axis each factor has absolute value at most one. For any N , bounding the first N factors by their reciprocal arguments shows that $H(t) = O_N(|t|^{-N})$ as $|t| \rightarrow \infty$. Fourier inversion thus gives a C^∞ density u of the law of Y . It is even, nonnegative, has integral one, and vanishes outside $[-1, 1]$. Its compact support and smoothness imply that H and all its derivatives are Schwartz functions on $\mathbb{R}$.

The first summand in (2.1) is uniform on $[-\frac{1}{2}, \frac{1}{2}]$, with density one. The remaining sum has support in the same interval and has no endpoint atoms. Convolution therefore gives $u(0) = 1$. This is the conventional normalization used in the cited Rvachev-function literature and in the repository.

Lemma 2.1 (Complete real zero set and multiplicity). *The real zeros of H are exactly the nonzero integers. For $n \in \mathbb{Z} \setminus \{0\}$,*

$$\text{ord}_n H = \nu_2(|n|) + 1. \quad (2.3)$$

In particular, $H(a/2) \neq 0$ for every odd integer a .

Proof. A factor in (2.2) vanishes precisely when t is a nonzero multiple of 2^j . If t is a nonzero integer of valuation d , exactly the factors $j = 0, \dots, d$ vanish, each simply. All other factors are nonzero. Their product is also nonzero: after finitely many factors, the sum of the absolute deviations from one converges and the factors stay in a neighborhood of one. The same nonvanishing argument applies when no factor vanishes. Thus no additional zeros arise from the infinite tail, and the multiplicity is $d + 1$. $\square$

The zero structure itself is classical; it is also visible in the canonical product in [1, eq. (9)]. We have given the short factor argument to make clear which exact property is used below.

2.2 Poisson summation in physical coordinates

Define the monomial defect

$$E_{r,M}(\theta) = Q_{M,\theta}(x^r) - I(x^r), \quad r \geq 0. \quad (2.4)$$

For a general polynomial, write $E_{P,M} = Q_{M,\theta}(P) - I(P)$ when the phase is understood.

Proposition 2.2 (Defect Fourier series). *For every $M > 0$ and integer $r \geq 0$,*

$$E_{r,M}(\theta) = \left(\frac{i}{2\pi}\right)^r \sum_{k \in \mathbb{Z} \setminus \{0\}} H^{(r)}(Mk) e^{2\pi i k \theta}. \quad (2.5)$$

The series and each of its phase derivatives converge absolutely and uniformly.

Proof. The function $f_r(x) = x^r u(x)$ is smooth and compactly supported. Its periodization is

$$g_r(\theta) = M^{-1} \sum_{j \in \mathbb{Z}} f_r((j + \theta)/M).$$

This has k th Fourier coefficient $\widehat{f}_r(Mk)$: substitute $x = (j + \theta)/M$ in the coefficient integral and join the resulting intervals. Its Fourier coefficients decrease faster than any power. Moreover,

$$H^{(r)}(t) = \int_{\mathbb{R}} (-2\pi i x)^r u(x) e^{-2\pi i t x} dx, \quad \widehat{f}_r(t) = \left(\frac{i}{2\pi}\right)^r H^{(r)}(t).$$

Removing the zero coefficient, which equals $I(x^r)$, proves (2.5). Rapid decay also proves all the stated convergence claims. $\square$

Corollary 2.3 (Uniform base order at an integer mesh). *If $M \in \mathbb{N}_{>0}$ and $d = v_2(M)$, then $Q_{M,\theta} = I$ on $\mathcal{P}_d$ at every phase. This uniform order is exactly d .*

Proof. Every nonzero integer Mk has zero multiplicity at least $d + 1$ in H . Thus all the terms in (2.5) vanish for $r \leq d$. For $r = d + 1$, the coefficient at $k = 1$ is nonzero by Lemma 2.1, so the defect cannot vanish identically. $\square$

The repository's unnormalized lattice-coordinate defect uses $(k + \theta)^r u((k + \theta)/M)$ and the corresponding integral. It is $M^{r+1} E_{r,M}(\theta)$. This observation reconciles (2.5) with the source-module statement and prevents an otherwise easy normalization error.

3 An odd-multiplier inequality and the first alias

Lemma 3.1 (Integer-multiple sine bound). *For real z and positive integer k ,*

$$|\sin(kz)| \leq k |\sin z|. \quad (3.1)$$

For odd k , also $|\cos(kz)| \leq k |\cos z|$.

Proof. The identity

$$e^{ikz} - e^{-ikz} = (e^{iz} - e^{-iz}) \sum_{j=0}^{k-1} e^{i(k-1-2j)z}$$

and the triangle inequality prove the sine estimate. For odd k , replace z by $z + \pi/2$ and take absolute values to obtain the cosine estimate. The argument includes the points where the right side is zero. $\square$

Lemma 3.2 (Odd-multiplier domination). *If a, k are positive odd integers, then*

$$\frac{|H(ak/2)|}{|H(a/2)|} \leq \frac{1}{k^2}. \quad (3.2)$$

Proof. For each factor of (2.2), put $z_j = \pi a/2^{j+1}$. None of $\sin z_j$ is zero. By Lemma 3.1,

$$\frac{|\operatorname{sinc}(kz_j)|}{|\operatorname{sinc}(z_j)|} = \frac{|\sin(kz_j)|}{k|\sin z_j|} \leq 1.$$

The first two factors are stronger. Oddness gives

$$|\sin(\pi ak/2)| = |\sin(\pi a/2)| = 1, \quad |\sin(\pi ak/4)| = |\sin(\pi a/4)| = \frac{1}{\sqrt{2}}.$$

Their ratios are therefore exactly $1/k$ each. For every finite product containing those two factors, the full ratio is at most k^{-2} . Pass to the limit, using $H(a/2) \neq 0$ from Lemma 2.1. $\square$

Remark 3.3. A generic rapid-decay estimate bounds $H(ak/2)$ by an unspecified constant that may be much larger than $|H(a/2)|$. Such an estimate cannot rule out extra zeros when the first coefficient is small. The content of (3.2) is its comparison with the *actual first coefficient*, uniformly over every odd a .

Lemma 3.4 (Exact first nonzero alias). *Let $M = 2^d a$ with a positive odd, and put $p = d + 1$. For odd $k \in \mathbb{Z} \setminus \{0\}$,*

$$H^{(p)}(Mk) = -\frac{p!}{(Mk)^p} H(ak/2). \quad (3.3)$$

For nonzero even k , $H^{(p)}(Mk) = 0$.

Proof. Write $n = Mk$. For odd k , separate the first p factors:

$$H(t) = \left(\prod_{j=0}^d \operatorname{sinc}(\pi t/2^j) \right) H(t/2^{d+1}).$$

At $t = n$, each of the displayed factors has a simple zero, with derivative $(-1)^{n/2^j}/n$. In the p th derivative, only the term differentiating every one of these factors once survives. Its combinatorial coefficient is $p!$. Its sign is

$$(-1)^{\sum_{j=0}^d n/2^j} = (-1)^{ak(2^{d+1}-1)} = -1.$$

The remaining factor is $H(ak/2)$, giving (3.3). The argument is valid for negative odd k too. For even k , Lemma 2.1 gives multiplicity at least $p + 1$. $\square$

Define the nonzero real number and the reference harmonic

$$B_M = -\frac{2p!}{(2\pi M)^p} H(a/2), \quad A_M = |B_M|, \quad \tau_p(\theta) = \cos(2\pi\theta + p\pi/2). \quad (3.4)$$

Pairing k and $-k$ in (2.5) now yields the exact real series

$$\frac{E_{p,M}(\theta)}{B_M} = \sum_{\substack{k \geq 1 \\ k \text{ odd}}} \frac{H(ak/2)}{H(a/2) k^p} \cos(2\pi k\theta + p\pi/2). \quad (3.5)$$

The sign in B_M and the phase shift $p\pi/2$ should both be retained; replacing this display by an unsigned amplitude would lose the defect's orientation.

4 The complete phase theorem and its stability estimates

For $p \geq 1$, put

$$\delta_p = \sum_{\substack{k \geq 3 \\ k \text{ odd}}} k^{-(p+1)} = (1 - 2^{-(p+1)})\zeta(p+1) - 1. \quad (4.1)$$

Then $0 < \delta_p \leq \delta_1 = \pi^2/8 - 1 < 1$. No evaluation of ζ is needed for the strict inequality: termwise comparison with $1/(k(k-1))$ gives

$$\delta_p \leq \sum_{\substack{k \geq 3 \\ k \text{ odd}}} \frac{1}{k^2} < \sum_{k=3}^{\infty} \frac{1}{k(k-1)} = \frac{1}{2}. \quad (4.2)$$

This weaker rational bound is sufficient for the classification.

Theorem 4.1 (Complete first-superconvergent phases). *Let $M \in \mathbb{N}_{>0}$, $d = v_2(M)$, $p = d + 1$, and $a = M/2^d$. For every $\theta \in \mathbb{T}$,*

$$\left| \frac{E_{p,M}(\theta)}{B_M} - \tau_p(\theta) \right| \leq \delta_p |\tau_p(\theta)|. \quad (4.3)$$

Consequently $Q_{M,\theta} = I$ on $\mathcal{P}_p$ if and only if

$$\theta \in Z_p := \begin{cases} \{0, \frac{1}{2}\} & (\text{mod } 1), \quad p \text{ odd}, \\ \{\frac{1}{4}, \frac{3}{4}\} & (\text{mod } 1), \quad p \text{ even}. \end{cases} \quad (4.4)$$

There are exactly two such phases modulo one. Away from them, $E_{p,M}/B_M$ has the same sign as τ_p .

Proof. The $k = 1$ term in (3.5) is τ_p . For odd k , Lemma 3.1 implies

$$|\cos(2\pi k\theta + p\pi/2)| \leq k |\cos(2\pi\theta + p\pi/2)|.$$

Indeed, for even p this is the cosine inequality, and for odd p it is the sine inequality, with harmless signs. By Lemma 3.2, the remaining terms have total absolute value at most

$$\sum_{\substack{k \geq 3 \\ k \text{ odd}}} \frac{1}{k^{p+2}} k |\tau_p(\theta)| = \delta_p |\tau_p(\theta)|.$$

This proves (4.3). If $\tau_p(\theta) = 0$, the bound forces the defect to vanish. If $\tau_p(\theta) \neq 0$, the ratio of $E_{p,M}(\theta)/B_M$ to $\tau_p(\theta)$ lies in $[1 - \delta_p, 1 + \delta_p] \subset (0, \infty)$, so the defect is nonzero and has the asserted sign. The zeros of τ_p are exactly (4.4). Finally, all lower monomial defects vanish by Corollary 2.3; exactness on $\mathcal{P}_p$ is therefore equivalent to vanishing of this one defect. $\square$

Corollary 4.2 (Simplicity, global phase conditioning, and polynomial errors). *Under the hypotheses of Theorem 4.1,*

$$(1 - \delta_p)A_M |\tau_p(\theta)| \leq |E_{p,M}(\theta)| \leq (1 + \delta_p)A_M |\tau_p(\theta)|. \quad (4.5)$$

Every $\theta_0 \in Z_p$ is a simple zero, and

$$2\pi A_M (1 - \delta_p) \leq |E'_{p,M}(\theta_0)| \leq 2\pi A_M (1 + \delta_p). \quad (4.6)$$

For $\rho = \text{dist}_{\mathbb{T}}(\theta, Z_p) \in [0, \frac{1}{4}]$,

$$4(1 - \delta_p)A_M \rho \leq |E_{p,M}(\theta)| \leq 2\pi(1 + \delta_p)A_M \rho. \quad (4.7)$$

If $P(x) = c_p x^p + \dots + c_0$, then $Q_{M,\theta}(P) - I(P) = c_p E_{p,M}(\theta)$.

Proof. The first assertion follows immediately from (4.3). Termwise differentiation is justified either by Proposition 2.2 or by

$$\sum_{\substack{k \geq 3 \\ k \text{ odd}}} k \left| \frac{H(ak/2)}{H(a/2)k^p} \right| \leq \delta_p.$$

It gives the global estimate

$$\left| (E_{p,M}/B_M)'(\theta) - \tau_p'(\theta) \right| \leq 2\pi\delta_p. \quad (4.8)$$

At a zero of τ_p , the reference derivative has absolute value 2π , proving (4.6) and simplicity. Next, $|\tau_p(\theta)| = \sin(2\pi\rho)$. Concavity of sine on $[0, \pi/2]$ and the bound $\sin t \leq t$ give $4\rho \leq \sin(2\pi\rho) \leq 2\pi\rho$. Combine this with (4.5). The polynomial statement uses the vanishing of every lower monomial defect. $\square$

4.1 Binary orientation and a uniform first-harmonic limit

The sign of the amplitude can be computed without evaluating an infinite product. Let $s_2(a)$ be the number of ones in the binary expansion of a . For positive odd a ,

$$\operatorname{sgn} H(a/2) = (-1)^{s_2(a)-1}, \quad \operatorname{sgn} B_M = (-1)^{s_2(a)}. \quad (4.9)$$

To check this, the sign of the j th sine factor is $(-1)^{\lfloor a/2^{j+1} \rfloor}$, and

$$\sum_{j \geq 0} \left\lfloor \frac{a}{2^{j+1}} \right\rfloor = a - s_2(a).$$

The latter identity follows by expanding a in binary and summing the contribution $2^m - 1$ of each nonzero digit in position m . Since a is odd, the claimed parity follows. This digital sign law is part of the classical Fourier description; here it gives the orientation of the new zero classification.

As $p \rightarrow \infty$, $\delta_p \rightarrow 0$ by dominated convergence. Equations (4.3) and (4.8) imply a C^1 first-harmonic limit uniform over *all* odd parts a of the mesh. This is a relative statement after normalization by B_M . It does not say that the absolute amplitudes A_M are bounded below uniformly in M or a .

Table 1: Uniform relative-tail bounds. The bound is independent of the odd part of the mesh.

$p = v_2(M) + 1$	δ_p	Guaranteed factor $1 - \delta_p$
1	0.233700550136	0.766299449864
2	0.051799790265	0.948200209735
3	0.014678031604	0.985321968396
4	0.004523762795	0.995476237205
5	0.001447076641	0.998552923359

5 All translation-uniform phase filters

5.1 Definition and a Fourier criterion

Let μ be a finite complex Borel measure on $\mathbb{T}$ with $\mu(\mathbb{T}) = 1$. Its positive-sign Fourier coefficients are

$$c_\mu(k) = \int_{\mathbb{T}} e^{2\pi i k \phi} d\mu(\phi). \quad (5.1)$$

Define the filtered quadrature

$$Q_{M,\theta}^\mu(P) = \int_{\mathbb{T}} Q_{M,\theta+\phi}(P) d\mu(\phi). \quad (5.2)$$

The atomic case $\mu = \sum_{j=1}^N w_j \delta_{\phi_j}$ is a weighted average of N translated quadratures, with $\sum_j w_j = 1$. The weights need not be positive or real.

Definition 5.1 (Uniform exactness). The filter μ is uniformly exact through degree r at mesh M if

$$Q_{M,\theta}^\mu(P) = I(P) \quad \text{for every } \theta \in \mathbb{T} \text{ and every } P \in \mathcal{P}_r.$$

The filter is fixed as θ varies. It cannot adapt its weights or relative phases to θ .

Lemma 5.2 (Spectral exactness criterion). *Uniform exactness through degree r is equivalent to*

$$c_\mu(k)H^{(j)}(Mk) = 0 \quad (k \neq 0, 0 \leq j \leq r). \quad (5.3)$$

Proof. Integrate (2.5) against μ . Absolute uniform convergence and finite total variation justify interchanging integration and summation, giving

$$Q_{M,\theta}^\mu(x^j) - I(x^j) = \left(\frac{i}{2\pi}\right)^j \sum_{k \neq 0} c_\mu(k)H^{(j)}(Mk)e^{2\pi i k \theta}.$$

A continuous periodic function vanishes identically if and only if every one of its Fourier coefficients vanishes. Apply this to each monomial and use linearity. $\square$

We use $T_h\mu$ for the pushforward of μ under $\phi \mapsto \phi + h$, and write

$$U_L = \frac{1}{L} \sum_{j=0}^{L-1} \delta_{j/L}, \quad L \in \mathbb{N}_{>0}. \quad (5.4)$$

Haar measure $m_{\mathbb{T}}$ means normalized Lebesgue measure on the phase circle.

Lemma 5.3 (Cyclic invariance and Fourier support). *For a finite complex measure μ and integer $L \geq 1$, the following are equivalent:*

$$T_{1/L}\mu = \mu; \quad c_\mu(k) = 0 \text{ whenever } L \nmid k; \quad \mu * U_L = \mu. \quad (5.5)$$

Proof. Translation multiplies $c_\mu(k)$ by $e^{2\pi i k/L}$, and $c_{U_L}(k)$ equals 1 for $L \mid k$ and 0 otherwise. Thus all three conditions give the same Fourier coefficients. Fourier coefficients uniquely determine a finite complex measure: equality on trigonometric polynomials extends to equality on continuous functions by uniform approximation, and continuous functions determine such measures. $\square$

5.2 Arithmetic rigidity for every real mesh

Theorem 5.4 (Complete phase-measure classification). *Let $M > 0$, $r \geq 0$, and let μ be a mass-one finite complex Borel measure on $\mathbb{T}$.*

(a) *If $M = a/b$ is rational in lowest terms, with a, b positive, put*

$$d = v_2(a), \quad s = \max(0, r - d), \quad L = b2^s. \quad (5.6)$$

Then μ is uniformly exact through degree r if and only if

$$T_{1/L}\mu = \mu. \quad (5.7)$$

*Equivalently, $\mu = \mu * U_L$, or its Fourier coefficients vanish outside $L\mathbb{Z}$.*

(b) *If M is irrational, uniform exactness even for constants holds if and only if $\mu = m_{\mathbb{T}}$. Haar measure is exact for every polynomial degree, for every real $M > 0$.*

Proof. Suppose first that $M = a/b$ is reduced. If $b \nmid k$, then Mk is not an integer, so $H(Mk) \neq 0$ by Lemma 2.1. The $j = 0$ condition in Lemma 5.2 forces $c_\mu(k) = 0$.

For $k = b\ell \neq 0$, the point $Mk = a\ell$ is an integer and its zero multiplicity is

$$\text{ord}_{a\ell} H = d + v_2(|\ell|) + 1.$$

All derivatives through degree r vanish there exactly when this multiplicity is greater than r . If it is at most r , the derivative at the multiplicity is nonzero and forces $c_\mu(k) = 0$. Thus the frequencies on which $c_\mu(k)$ is permitted to be nonzero are exactly

$$k \in b 2^{\max(0, r-d)} \mathbb{Z} = L\mathbb{Z}.$$

The spectral criterion and Lemma 5.3 prove both directions of part (a), including the case $r \leq d$.

If M is irrational, Mk is nonintegral for every nonzero integer k . Constant exactness therefore forces $c_\mu(k) = 0$ for every $k \neq 0$. Together with $c_\mu(0) = 1$, these are precisely the Fourier coefficients of Haar measure, so uniqueness gives $\mu = m_{\mathbb{T}}$. Conversely, Haar averaging suppresses all nonzero frequencies. Alternatively, directly for any compactly supported integrable f ,

$$\int_0^1 \frac{1}{M} \sum_{k \in \mathbb{Z}} f\left(\frac{k + \theta + \phi}{M}\right) d\phi = \int_{\mathbb{R}} f(x) dx,$$

because the transformed integration intervals partition $\mathbb{R}$. Take $f = Pu$ to obtain the final assertion. $\square$

Corollary 5.5 (No finite irrational-mesh filter; all-order uniqueness). *No mass-one finitely atomic phase filter at an irrational mesh can be uniformly exact even on constants. For any fixed real $M > 0$, the only mass-one finite complex phase measure uniformly exact through every polynomial degree is Haar measure.*

Proof. Haar measure is non-atomic, proving the first statement. At a rational mesh, every fixed nonzero frequency eventually lies outside $b2^{\max(0, r-d)}\mathbb{Z}$ as $r \rightarrow \infty$. Exactness at all orders forces all nonzero Fourier coefficients to vanish. The irrational case is already part of Theorem 5.4. $\square$

The obstruction in Corollary 5.5 is about an exact identity for all translations. It does not preclude high numerical accuracy at an irrational mesh, or exactness at one specially chosen translation.

6 Sharp phase counts, equality cases, and refinement

6.1 Orbit structure, including signed and complex weights

By an *active phase* of an atomic measure we mean a point with a nonzero weight, after coincident phases have been coalesced and zero weights removed. Counts below always refer to distinct active phases.

Proposition 6.1 (Atomic orbit classification). *A finitely atomic measure invariant under $T_{1/L}$ is a finite union of complete cosets of the cyclic subgroup $\{0, 1/L, \dots, (L-1)/L\}$, with a constant weight on each coset. Its number N of active phases is a positive multiple of L if its total mass is one. In particular $N \geq L$. If $N = L$ and the mass is one, the measure is exactly $T_\alpha U_L$ for some $\alpha \in \mathbb{T}$.*

Proof. The translation by $1/L$ has orbits of length exactly L on $\mathbb{T}$. Invariance of the measure implies that the atom at $x + 1/L$ has the same weight as the atom at x . A nonzero atom therefore generates a full orbit of L nonzero atoms of equal weight. Distinct orbits are disjoint. Conversely, every such union is invariant. If there is only one orbit, its weight w satisfies $Lw = 1$, so $w = 1/L$. $\square$

Corollary 6.2 (Minimum phases at a prescribed order). *At a reduced rational mesh a/b , uniform exactness through degree r requires at least*

$$L_r = b2^{\max(0, r - \nu_2(a))} \quad (6.1)$$

active phases. This number is sufficient. Every filter attaining this minimum is a translate of U_{L_r} , even when arbitrary complex weights are allowed.

Proof. Apply Theorem 5.4 and Proposition 6.1. The filter U_{L_r} attains the bound, and the orbit equality case is unique up to translation. $\square$

For $r < \nu_2(a)$ a minimum-size filter automatically has additional exactness through degree $\nu_2(a)$. The phrase “through degree r ” does not require failure at degree $r + 1$.

6.2 Exactly N phases versus a budget of at most N

Theorem 6.3 (Optimal order with exactly N active phases). *Fix $M = a/b > 0$ in lowest terms and $N \geq 1$. A mass-one filter with exactly N active phases can be uniformly exact on constants if and only if $b \mid N$. When $b \mid N$, its largest achievable uniform degree is*

$$r_{=N} = \nu_2(a) + \nu_2(N/b). \quad (6.2)$$

The uniform grid $T_\alpha U_N$ attains this degree. More generally, every exact- N maximizer consists of $N/(b2^{\nu_2(N/b)})$ complete cosets of the subgroup of order $b2^{\nu_2(N/b)}$, with constant nonzero weights on the cosets and total mass one.

Proof. Constant exactness requires invariance under $1/b$, hence $b \mid N$. For any attainable degree $r > \nu_2(a)$, Theorem 5.4 and Proposition 6.1 force $b2^{r - \nu_2(a)} \mid N$, giving the upper bound (6.2). This is also an upper bound when compared with orders at most $\nu_2(a)$. If $b \mid N$, the uniform grid U_N is invariant under every cyclic translation whose order divides N , so it attains the bound. At that order the orbit description follows from Proposition 6.1. The number of these minimal orbits is odd; equivalently, the next required orbit length does not divide N , so no such exact- N filter can attain the next degree. $\square$

Theorem 6.4 (Optimal order under a phase budget). *Fix a reduced rational mesh $M = a/b > 0$ and a budget $N \geq 1$. If $N < b$, no mass-one atomic filter within budget is uniformly exact on constants. If $N \geq b$, put*

$$s_N = \lfloor \log_2(N/b) \rfloor, \quad L_* = b2^{s_N}. \quad (6.3)$$

The maximum attainable uniform degree and all its optimizers are

$$r_{\leq N} = \nu_2(a) + s_N, \quad \mu = T_\alpha U_{L_*} \quad (\alpha \in \mathbb{T}). \quad (6.4)$$

Every optimizer uses exactly L_ active phases, possibly leaving part of the budget unused. In particular all its weights are positive and equal.*

Proof. The minimum-size formula (6.1) gives the upper bound, and U_{L_*} attains it. An optimizer must be invariant under $1/L_*$, so its active-phase count is a positive multiple of L_* . But $L_* \leq N < 2L_*$. Its count is therefore exactly L_* , and Proposition 6.1 forces a single uniformly weighted coset. $\square$

Example 6.5 (Four phases outperform a uniform six-phase average). At $M = 12$ one has $\nu_2(M) = 2$. A budget of six phases allows degree $2 + \lfloor \log_2 6 \rfloor = 4$, attained by a translated four-phase grid. A uniform six-phase grid has degree only $2 + \nu_2(6) = 3$. In fact, no filter with *exactly* six active phases, regardless of signed or complex weights, can reach uniform degree four. There is no paradox: the four-phase filter is not an exact-six-phase filter. Adding two genuinely active phases destroys its required four-cycle orbit structure.

Table 2: Examples of the sharp at-most- N budget theorem. “None” means that uniform constant exactness is impossible.

Mesh parameter M	Budget N	Optimal uniform degree	Phases used
12	4	4	4
12	6	4	4
3/5	4	None	—
3/5	5	0	5
3/5	12	1	10
8/3	13	5	12

6.3 The equality filters are exact lattice refinements

Proposition 6.6 (Refinement identity). *For every $M > 0$, integer $L \geq 1$, and phase offset α ,*

$$\frac{1}{L} \sum_{j=0}^{L-1} Q_{M, \theta + \alpha + j/L}(P) = Q_{LM, L(\theta + \alpha)}(P). \quad (6.5)$$

The identity holds for any function for which the displayed sums make sense, not only polynomials.

Proof. The left-hand nodes are $(k + \theta + \alpha + j/L)/M = (Lk + j + L(\theta + \alpha))/(LM)$, each with weight $1/(LM)$. The map $(k, j) \mapsto Lk + j$ is a bijection from $\mathbb{Z} \times \{0, \dots, L-1\}$ onto $\mathbb{Z}$. $\square$

The identity proves that the budget-optimal filters are not exotic cancellation schemes. They simply realize the most efficient admissible refinement of the original lattice.

Corollary 6.7 (Complete extra-degree phases of a minimum refinement). *Let $M = a/b$ be reduced, let $d = v_2(a)$, choose $s \geq 0$, and put $L = b2^s$ and $r = d + s$. For $\mu = T_\alpha U_L$, the filtered rule is uniformly exact through degree r . It is exact through degree $r + 1$ at a particular translating phase θ if and only if*

$$L(\theta + \alpha) \in \begin{cases} \{0, \frac{1}{2}\} \pmod{1}, & r \text{ even}, \\ \{\frac{1}{4}, \frac{3}{4}\} \pmod{1}, & r \text{ odd}. \end{cases} \quad (6.6)$$

There are exactly $2L$ such translating phases modulo one.

Proof. The refined mesh $LM = a2^s$ is a positive integer of valuation r . Apply Proposition 6.6 and Theorem 4.1. Each of the two refined phases has exactly L preimages under multiplication by L on $\mathbb{T}$. $\square$

For example, $M = 3/5$ with the minimum ten-phase filter has uniform degree one. It has degree at least two exactly at the twenty translations satisfying $10(\theta + \alpha) \in \{1/4, 3/4\} \pmod{1}$.

7 Finite recognition and exact filter certificates

The phase-measure theorem uses infinitely many Fourier coefficients. For an atomic filter with a known support bound, finitely many suffice. No positivity assumption is needed.

Theorem 7.1 (Finite recognition of cyclic invariance). *Let μ be a real signed measure supported on at most $N \geq 1$ points of $\mathbb{T}$, and let $L \geq 1$. Then*

$$T_{1/L} \mu = \mu \iff c_\mu(k) = 0 \quad (1 \leq k \leq N, L \nmid k). \quad (7.1)$$

For a complex measure supported on at most N points, the sufficient and necessary finite test is instead

$$c_\mu(k) = 0 \quad (1 \leq k \leq 2N - 1, L \nmid k). \quad (7.2)$$

Proof. Necessity follows from Lemma 5.3. For sufficiency in the real case, let $\eta = T_{1/L}\mu - \mu$. It is real signed, has mass zero, and is supported on at most $2N$ points. Its Fourier coefficients are

$$c_\eta(k) = (e^{2\pi i k/L} - 1)c_\mu(k).$$

The assumed moments make these vanish for $1 \leq k \leq N$: when $L \mid k$, the prefactor is zero, and otherwise the assumed coefficient is zero. The zero mode vanishes automatically. Since η is real, $c_\eta(-k) = \overline{c_\eta(k)}$, so all modes from $-N$ through N vanish.

Write $\eta = \sum_{j=1}^S v_j \delta_{x_j}$ with distinct support points and $S \leq 2N$, and put $z_j = e^{2\pi i x_j}$. If $S = 0$ there is nothing to prove. Otherwise the S equations for $k = -N, -N+1, \dots, -N+S-1$ give

$$\sum_{j=1}^S (v_j z_j^{-N}) z_j^\ell = 0, \quad \ell = 0, \dots, S-1.$$

Their Vandermonde determinant is $\prod_{i < j} (z_j - z_i) \neq 0$. Thus every $v_j = 0$ and $\eta = 0$.

For complex μ , no conjugacy of negative and positive coefficients may be assumed. The test (7.2) gives vanishing of $c_\eta(k)$ for $k = 0, \dots, 2N-1$. The first S of these equations form an ordinary Vandermonde system, again giving $\eta = 0$. $\square$

Remark 7.2 (Why the support bound and realness matter). Without a support bound, a measure can have any prescribed finite initial set of nonzero Fourier modes equal to zero and still fail the invariance condition. For example, a uniform grid of sufficiently large order does so. Even within the real N -point class, the upper index N cannot be universally reduced to $N-1$: whenever $L \nmid N$, the measure U_N passes every test below N but is not invariant under $1/L$. For complex weights, the real-case proof cannot silently use $c_\mu(-k) = \overline{c_\mu(k)}$.

7.1 Two exact certificate formats

For rational phases and rational weights, an *orbit certificate* is particularly simple. Coalesce phases modulo one, remove zero weights, translate every active phase by $1/L$, and compare the resulting weighted finite maps exactly. This requires only rational arithmetic. The supplied Python implementation uses this route; it does not approximate roots of unity to certify a zero.

For algebraic unit-circle coordinates $z_j = e^{2\pi i \phi_j}$ and algebraic weights, a *moment certificate* can instead verify the finite identities in Theorem 7.1 in an exact algebraic number field. The theorem then connects finitely many algebraic identities to every translating phase and every polynomial through the specified degree. Efficient discovery of the certificate and verification of its identities are separate algorithmic tasks.

The real-case proof has a finite, transparent trust boundary: a support bound, the required Fourier moments, and nonvanishing of a Vandermonde determinant. It does not require experimentally testing many translated quadratures.

8 Transfer to moment-deconvolved polynomial synthesis

Define a linear operator on polynomials by

$$(\mathcal{T}A)(t) = \int_{\mathbb{R}} A(t-x)u(x) dx. \quad (8.1)$$

For a polynomial of degree n , the leading term is unchanged because $\int u = 1$. Thus $\mathcal{T}$ is triangular with diagonal one on every $\mathcal{P}_r$, and consequently is an automorphism of that space. For each polynomial P , let $A_P = \mathcal{T}^{-1}P$. Define

$$(S_{M,\theta}P)(t) = \frac{1}{M} \sum_{k \in \mathbb{Z}} A_P\left(t - \frac{k+\theta}{M}\right) u\left(\frac{k+\theta}{M}\right). \quad (8.2)$$

The filtered version is $\mathcal{S}_{M,\theta}^\mu P = \int_{\mathbb{T}} \mathcal{S}_{M,\theta+\phi} P d\mu(\phi)$.

Theorem 8.1 (Exactness and synthesis are equivalent). *For a fixed M , phase θ , and degree bound r , the following are equivalent:*

$$\mathcal{Q}_{M,\theta} = I \text{ on } \mathcal{P}_r; \quad \mathcal{S}_{M,\theta} P = P \text{ for every } P \in \mathcal{P}_r.$$

The same equivalence holds for the filtered operators with any mass-one finite complex phase measure. It holds both at a fixed phase and uniformly in phase. Therefore all preceding classification, phase-count, and extra-degree phase theorems transfer to these synthesis operators.

Proof. Assume quadrature exactness. For each fixed t , the function $x \mapsto A_P(t-x)$ is a polynomial of degree at most r . Exactness therefore gives

$$(\mathcal{S}_{M,\theta} P)(t) = \int_{\mathbb{R}} A_P(t-x) u(x) dx = P(t).$$

Conversely, given $B \in \mathcal{P}_r$, choose $A(t) = B(-t)$ and $P = \mathcal{T}A$. Then $A_P = A$, and synthesis evaluated at $t = 0$ gives

$$\mathcal{Q}_{M,\theta}(B) = (\mathcal{S}_{M,\theta} P)(0) = P(0) = \int_{\mathbb{R}} B(x) u(x) dx.$$

The filtered proof is identical, replacing the quadrature functional by its phase average. Quantifying over θ preserves the equivalence. $\square$

8.1 Appell polynomials and concrete examples

Put $A_n = \mathcal{T}^{-1}(t^n)$. With $m_n = \int x^n u(x) dx$, the first moments are

$$m_0 = 1, \quad m_1 = m_3 = 0, \quad m_2 = \frac{1}{9}, \quad m_4 = \frac{19}{675}.$$

Triangular inversion gives

$$\begin{aligned} A_0(t) &= 1, & A_1(t) &= t, & A_2(t) &= t^2 - \frac{1}{9}, \\ A_3(t) &= t^3 - \frac{1}{3}t, & A_4(t) &= t^4 - \frac{2}{3}t^2 + \frac{31}{675}. \end{aligned}$$

These identities can be checked simply by applying $\mathcal{T}$. The exponential generating identity is

$$\sum_{n \geq 0} A_n(t) \frac{z^n}{n!} = \frac{e^{tz}}{\int_{\mathbb{R}} e^{zx} u(x) dx}, \quad (8.3)$$

understood either as a formal power series or locally near $z = 0$. Evenness of u removes a possible minus sign in the denominator. We do not assert that the reciprocal moment-generating function is entire.

For an integer mesh M with $p = \nu_2(M) + 1$, the leading term of A_p is t^p . Hence the degree- p synthesis error has the particularly simple form

$$(\mathcal{S}_{M,\theta} t^p)(t) - t^p = (-1)^p E_{p,M}(\theta), \quad (8.4)$$

independent of t . Indeed, the integrand $A_p(t-x)$ has leading coefficient $(-1)^p$ as a polynomial in x , and every lower quadrature defect vanishes. For a general $P \in \mathcal{P}_p$, multiply the right side by its coefficient of t^p .

This gives more than a correspondence between two zero sets: the same quantitative bounds control the full leading synthesis error.

9 Independent exact and numerical checks

The accompanying program `reproducibility/verify.py` is standalone; it imports no repository code. Its exact component uses Python rational arithmetic. Its numerical component uses 65-decimal-digit `mpmath` arithmetic and is explicitly not an interval or kernel-checked proof. The recorded output is in `validation/results.json`.

9.1 Exact dyadic evaluation and moments

For the bounded Fabius function on $[0, 1]$, use $u(x) = F(1 - |x|)$ on $[-1, 1]$. The dyadic evaluation identities are classical; see [2, eqs. (19), (22), and Prop. 10]. The implemented normalization is

$$d_n = n!2^{n(n-1)/2}F(2^{-n}), \quad d_0 = 1, \quad (9.1)$$

with the recurrence

$$(n+1)(2^n - 1)d_n = \sum_{k=0}^{n-1} \binom{n+1}{k} d_k \quad (n \geq 1). \quad (9.2)$$

For a dyadic $x \in (0, 1)$, choose n so that $2^{-n} \leq x < 2^{1-n}$, and put $y = x - 2^{-n}$. Then

$$F(x) = -F(y) + \sum_{k=0}^n \frac{2^{k(k+1)/2}}{k!} F(2^{k-n}) y^k. \quad (9.3)$$

Each recursion removes the leading nonzero binary digit, so exact evaluation terminates. In particular $F(1/2) = 1/2$ and $F(1/4) = 5/72$. These evaluation identities are used only for supplemental sample checks; the proofs of the main theorems do not depend on them.

Moments are computed independently from the distributional identity $Y \stackrel{d}{=} (V + Y')/2$, where Y' has the law of Y and is independent of $V \sim \text{Unif}[-1, 1]$. Odd moments vanish, and for even $n > 0$,

$$(2^n - 1)m_n = \sum_{\substack{0 \leq k < n \\ k \text{ even}}} \binom{n}{k} \frac{m_k}{n - k + 1}. \quad (9.4)$$

This follows by expanding $(V + Y')^n$ and isolating the $k = n$ term.

Table 3: Exact dyadic checks. The last column checks one next-degree failure at each displayed mesh; it is not a general maximality theorem.

M	p	Chosen $\theta_0 \in Z_p$	$E_{p,M}(1/8)$	$E_{p+1,M}(\theta_0)$
1	1	0	35/288	-1/9
2	2	1/4	41543/4147200	17/1536
4	3	0	-52219/176947200	97/345600
8	4	1/4	-176635903/59929893273600	-9919/4529848320

The exact tests verify the lower-order vanishing and the threshold zero set on the 32-point phase grid for each $M \in \{1, 2, 4, 8, 16\}$. They also verify Fabius reflection on a 257-point grid, selected base values, and 216 lattice-refinement identities. Including the initial checks, the program records 1,118 exact function/quadrature/refinement checks and 1,026 exact finite-filter checks, for 2,144 exact checks in total.

9.2 Product comparisons and a numerical picture

The numerical tests evaluate 120 factors of (2.2) and 100 positive odd terms of (3.5). For odd $a = 1, 3, \dots, 63$ and odd $k = 3, 5, \dots, 63$, all 992 tests of (3.2) passed. The 32 binary-sign checks also passed. The largest observed value of $k^2|H(ak/2)/H(a/2)|$ in that finite test was approximately 0.750889. This is an observation, not a proposed optimal universal constant.

Comparison of the truncated Fourier series with exact dyadic quadratures on the test grid gave maximum absolute discrepancy approximately $2.65 \cdot 10^{-16}$. The number describes this specific finite computation; it is not a rigorous error certificate. The observed discrepancy is much larger than the arithmetic working precision, consistent with truncating the Fourier series rather than evaluating it exactly.

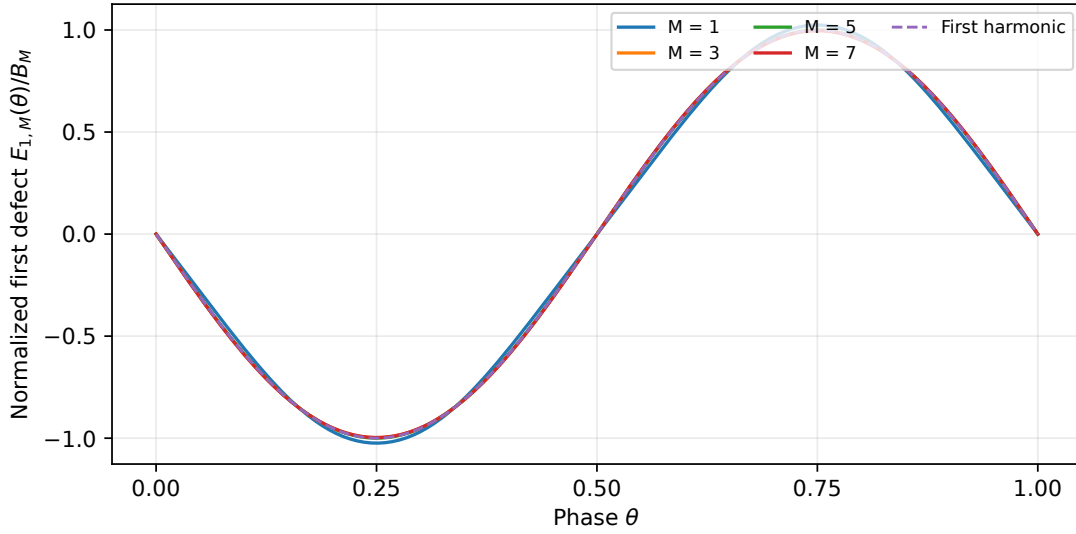


Figure 1: Normalized first defects for four odd mesh parameters, compared with the reference harmonic $-\sin(2\pi\theta)$. Theorem 4.1, not the plot, proves that there are no other zeros. The displayed curves use truncated Fourier series.

There is an elementary analytic truncation estimate for the product. If J is large enough that $|\pi t|2^{-J} \leq 1$, then all factors in the tail starting at J are positive and

$$0 \leq 1 - \prod_{j=J}^{\infty} \text{sinc}(\pi t/2^j) \leq \sum_{j=J}^{\infty} \frac{\pi^2 t^2}{6 \cdot 4^j} = \frac{2\pi^2 t^2}{9} 4^{-J}. \quad (9.5)$$

Here $1 - \prod(1 - a_j) \leq \sum a_j$ for $a_j \in [0, 1]$ is applied to $1 - \text{sinc}(\pi t/2^j)$. For the normalized first-defect series, truncation after an odd index K has absolute error at most $\sum_{k>K, k \text{ odd}} k^{-(p+2)}$, and relative error with respect to $|\tau_p(\theta)|$ at most $\sum_{k>K, k \text{ odd}} k^{-(p+1)}$. An interval implementation could combine these estimates with directed rounding. The supplied numerical program does not claim to do that.

10 Integration with the formal repository

10.1 Existing dependencies and new obligations

The inspected source already contains the Fourier-defect identity, valuation-based alias vanishing, and sufficient parity-phase theorems [4, 5]. In particular, the names

`summable_fourier_monomialRvachevSchwartz`

```
tsum_shifted_monomial_sub_integral
tsum_shifted_monomial_sub_integral_odd
tsum_shifted_polynomial_eq_integral_superconvergent
```

identify relevant existing declarations. The last theorem is a *sufficiency* theorem; its name alone should not be read as an if-and-only-if statement.

A formalization of the new necessity proof can be broken into three small analytic layers. First prove the sine multiple inequality and the two exact quarter/half-integer factor ratios. Next pass the finite-product ratio bound to the infinite product. Finally combine the odd defect series with the weighted tail estimate, using the rational bound $\delta_p < 1/2$ if a zeta-function evaluation would add unnecessary infrastructure. This yields completeness before the sharper constant is formalized.

The phase-filter classification requires a second development: Fourier coefficients of finite signed or complex measures on the circle, their behavior under translation, and Fourier uniqueness. The finite atomic case can be developed first, with direct finite sums, and then extended to arbitrary finite measures. The orbit-count and budget proofs are finite group-action arguments after the analytic exactness criterion.

The finite recognition theorem has an especially concrete formal route: represent the difference between a measure and its translate by a finite support list, establish its support-size bound, and use an invertible Vandermonde matrix on distinct unit-circle points. This must not use real conjugacy for complex weights; the two frequency ranges in Theorem 7.1 are genuinely different proof interfaces.

10.2 A proposed theorem boundary

For integer $M > 0$, a natural formal endpoint is the equivalence

$$(\forall P \in \mathcal{P}_{v_2(M)+1}, Q_{M,\theta}(P) = I(P)) \iff \theta \bmod 1 \in Z_{v_2(M)+1}.$$

The existing predicate uses explicit real representatives such as 0 and $1/2$. A new iff theorem must either work on the circle or explicitly reduce θ modulo one. Otherwise, an assertion phrased for arbitrary real θ would incorrectly exclude, for example, the phase 1.

For rational $M = a/b$, the second endpoint should expose the *reduced* denominator b , not an arbitrary representation of the rational number. At each degree it should connect uniform exactness to a cyclic-invariance predicate of order $b2^{\max(0, r-v_2(a))}$. The phase-count theorem must count active atoms after coalescing, not the length of a possibly redundant input list.

No new formal file or checked theorem is included in this delivery. The Python checks are not Lean certificates, and reading existing declarations is not the same as rebuilding and auditing their dependencies. The article's proofs stand independently as conventional proofs; formalization is a well-specified next step rather than a completed claim.

11 Further research questions and proposed projects

11.1 Fixed-phase maximality beyond the first extra degree

Question 1. For every integer $M > 0$ and every $\theta_0 \in Z_p$, $p = v_2(M) + 1$, is $E_{p+1,M}(\theta_0) \neq 0$? An affirmative answer would show that the selected phases gain exactly one degree, not merely classify the phases that gain at least one. The exact examples in Table 3 support this question but do not settle it. At the next degree, even aliases with valuation one also enter, and derivatives of the nonvanishing product tail occur at the odd aliases. The single first-harmonic estimate does not directly resolve the resulting competition.

11.2 Sharp domination constants and absolute conditioning

Question 2. Determine the best universal constant in the relative estimate (4.3), or the sharp value of

$$\sup_{\substack{a, k \geq 1 \text{ odd} \\ k \geq 3}} k^2 \frac{|H(ak/2)|}{|H(a/2)|}.$$

The proof gives an upper bound of one for the latter, but the finite experiment reaches only about 0.751. Can long binary patterns in a approach one, or is there a strict uniform improvement?

Question 3. Find effective upper and lower bounds on $A_M = 2p!|H(a/2)|/(2\pi M)^p$ in terms of the binary structure of a . The current theorem gives mesh-independent *relative* phase conditioning. An absolute error tolerance requires control of A_M , which can be very small. A useful result would separate the influence of large p from the small-divisor-like effects of the half-integer product.

11.3 Stability of approximate Fourier invariance

Question 4. Fix a support bound and a minimum separation between active phases. If the moments in Theorem 7.1 are only approximately zero, how close must the filter be to a union of equal-weight cyclic orbits? The Vandermonde proof suggests bounds involving the inverse Vandermonde norm. A meaningful theorem should state the topology, weight bounds, and separation assumptions explicitly: small finitely many moments alone do not imply small total-variation distance to an invariant measure.

Question 5. At meshes close to a rational a/b , quantify the loss of uniform exactness for the optimal rational filter. The exact rational/irrational dichotomy is discontinuous as a classification, while finite-degree numerical errors vary continuously with the mesh. A perturbation theorem could reconcile these two facts and give tolerances for implementation.

11.4 Other kernels, dimensions, and cost models

Question 6. Which other compactly supported infinite-convolution kernels admit a uniform first-coefficient ratio bound strong enough to classify every first-superconvergent phase? For base- q products, the nonzero-frequency support may split into several residue classes rather than just the odd integers. The zero multiplicities alone classify uniform filters, but do not by themselves classify special-phase zeros.

Question 7. For a product of Rvachev kernels in several variables, classify finite phase measures exact on total-degree polynomial spaces. A non-product filter may exploit aliases killed by a zero in *one* coordinate. Thus the one-dimensional invariance theorem does not simply imply that every optimal multidimensional filter is a product grid. The correct obstruction is a multivariate set of derivative indices and alias frequencies.

Question 8. How much can be saved when exactness is required at one fixed phase instead of every translating phase? The exponential phase-cost law proved here is tied to translation-uniformity. Allowing phase-dependent weights or independent node positions changes the problem and may permit substantially smaller formulas. A useful comparison would retain a clearly specified lattice-mixture constraint while dropping only the uniform-translation quantifier.

11.5 Approximation theory and formal completion

Question 9. Develop Peano-kernel or remainder estimates for nonpolynomial smooth integrands at the classified phases. The leading polynomial defect vanishes there, but an approximation theorem must also control higher derivatives and the dependence of the weights on the mesh. A sign-definite remainder formula would be particularly useful for certified numerical integration.

Question 10. Complete a Lean formalization of the necessity statement, then the atomic filter classification, then the general measure version. The first target is comparatively focused: it can reuse

the existing odd-alias series and needs the relative product bound plus an elementary convergent-series majorant. Formalizing the sharp zeta constant can be separated from formalizing the qualitative result.

12 Conclusion

The missing phase-completeness statement has a quantitative answer: the first defect cannot acquire additional zeros because its remaining odd harmonics are uniformly dominated *relative to* its first harmonic. The resulting theorem holds for every integer mesh parameter, including arbitrarily large odd parts.

At the level of translation-uniform phase mixtures, the answer is more rigid. The reduced rational denominator and the dyadic valuation of the numerator determine an exact cyclic symmetry group; that group determines all admissible filters and their sharp finite cost. Irrational meshes admit no finite filter even for uniform constant exactness. Minimum-size and budget-optimal filters are necessarily ordinary uniform lattice refinements, despite allowing signed or complex weights.

These results close a specifically documented repository gap and establish a broader classification around it. Fixed-phase maximality at the next degree, optimal constants, and multidimensional designs remain distinct, precisely formulated problems.

A A reusable first-harmonic zero-rigidity lemma

The core argument does not intrinsically depend on the Rvachev function. The following formulation isolates the transferable analytic mechanism.

Proposition A.1 (Odd-harmonic zero rigidity). *Let ε be either 0 or $\pi/2$, let $a_1 \in \mathbb{R} \setminus \{0\}$, and let $(a_k)_{k \geq 3, k \text{ odd}}$ be real coefficients such that*

$$\eta = \frac{1}{|a_1|} \sum_{\substack{k \geq 3 \\ k \text{ odd}}} k |a_k| < 1.$$

Then

$$f(x) = \sum_{\substack{k \geq 1 \\ k \text{ odd}}} a_k \cos(kx + \varepsilon)$$

has precisely the zeros of $\cos(x + \varepsilon)$, all simple, and

$$\left| \frac{f(x)}{a_1} - \cos(x + \varepsilon) \right| \leq \eta |\cos(x + \varepsilon)|.$$

At those zeros, $|a_1|(1 - \eta) \leq |f'(x)| \leq |a_1|(1 + \eta)$.

Proof. The weighted summability assumption gives uniform convergence of the series and its first derivative. For odd k , $|\cos(kx + \varepsilon)| \leq k |\cos(x + \varepsilon)|$ by Lemma 3.1. Summing the tail gives the displayed relative bound. Since $\eta < 1$, the sign and zero conclusions follow. At a reference zero, the first derivative of the first term has absolute value $|a_1|$, while the derivative tail has absolute value at most $\eta |a_1|$. $\square$

For Rvachev quadrature, the coefficients normalized by their first term satisfy $|a_k/a_1| \leq k^{-(p+2)}$, making $\eta \leq \delta_p$. The factor k in the weighted tail is not cosmetic. An unweighted small tail controls the defect away from a reference zero, but need not prevent additional zeros arbitrarily close to it.

B Reproducibility and the deliverable boundary

The archive contains this source, the compiled article, the figure assets, the independent checking program, a saved JSON result, and build instructions. From the archive root, run

```
python reproducibility/verify.py -plot
```

to regenerate the diagnostic output and figure. The program requires Python with `mpmath`; plotting additionally requires `numpy` and `matplotlib`. The exact arithmetic uses only the standard `fractions`, `math`, and `functools` modules.

Compile the article with

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The figure PDF is already included, so recompiling the article does not require rerunning the numerical program. No external bibliography processor is needed, and no font files are distributed.

The supplied exact checks validate finite identities and sample values. The supplied numerical checks validate neither an infinite product nor an infinite family of phases by themselves. The full statements are justified by the analytic proofs in the main text. No changes were made to the remote repository.

References

- [1] J. Arias de Reyna, *An infinitely differentiable function with compact support: Definition and properties*, English translation (2017) of *Definición y estudio de una función indefinidamente diferenciable de soporte compacto*, Rev. Real Acad. Ciencias Madrid **76** (1982), 21–38. <https://arxiv.org/abs/1702.05442>.
- [2] J. Arias de Reyna, *Arithmetic of the Fabius function*, arXiv:1702.06487, version 3 (2017). <https://arxiv.org/abs/1702.06487>.
- [3] V. G. Zakharov, *Polynomial spaces reproduced by elliptic scaling functions*, arXiv:1310.8425 (2013). <https://arxiv.org/abs/1310.8425>.
- [4] V. Reshetnikov and contributors, *ProveIt: RvachevSuperconvergentSynthesis.lean*, Analysis/FabiusFunction/Lean/FabiusFunction/, commit de4ac5ab1438bcbeed58481c4449b593074a1a1f. [Pinned source file on GitHub](#). The introductory documentation states sufficient selected phases and explicitly leaves their completeness unasserted.
- [5] V. Reshetnikov and contributors, *ProveIt: CombDefectSeries.lean*, Analysis/FabiusFunction/Lean/FabiusFunction/, commit de4ac5ab1438bcbeed58481c4449b593074a1a1f. [Pinned source file on GitHub](#). The source gives absolutely convergent defect series and odd-frequency support at the first threshold.